

SAMYUKTHA MA

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Professional Summary

AI & Data Analytics undergraduate with strong foundations in machine learning, data analysis, explainable AI, and production-oriented AI systems. Hands-on experience in ML validation, feature engineering, real time data visualization, dashboards, prompt engineering, and FinTech-style analytics across industrial, telecom, and aerospace research environments.

Education

Saveetha College of Liberal Arts and Sciences

2023--2026

BSc Artificial Intelligence

Skills

Programming & Analytics: Python, SQL (MS SQL Server), Advanced Excel, SPSS Machine Learning & AI: Supervised & Unsupervised Learning, Feature Engineering, Model Validation, Neural Networks, Random Forest, SVM, KNN, Generative AI
Prompt Engineering & LLMs: System prompt design, structured AI outputs, explainability-focused and risk-aware
Data Visualization & BI: Power BI, Interactive Dashboards, Data Cleaning & Transformation, Real-Time Visualizations
Web & UI: HTML, CSS, React (Functional Components), TypeScript.

Experience

SHAR SDSC-ISRO Range Operations Intern | Sriharikota | Dec 2025 – Jan 2026

- Improved operational monitoring reliability by developing real-time telemetry visualization workflows for range operations environments.
- Enhanced the accuracy and interpretability of high-frequency telemetry analysis through curve smoothing, interpolation, and preprocessing techniques.

Testing Analyst Intern Nokia Networks & Solutions | Jul 2025 – Aug 2025

- Increased dataset reliability and model consistency by validating ML datasets and identifying defects across telecom analytics pipelines.
- Reduced data quality issues impacting analytics performance through structured testing and defect documentation.
- Strengthened collaboration between testing and engineering teams, contributing to more stable and trustworthy ML outputs.

Production Planner & Industrial Operations Intern Alstom Transport India | May 2025 – Jun 2025

- Contributed to KPI improvement initiatives by analyzing production and operational datasets to identify workflow inefficiencies and performance gaps.
- Improved manufacturing performance visibility through time-study analysis, operational reporting, and data-driven process assessments.

Projects

Nova Spark – FinTech AI Market Intelligence Platform

- Built an AI-powered FinTech single-page application with a conversational assistant delivering explainable, risk-aware market analysis and simulated trade forecasting.
- Implemented technical indicators (RSI, MACD, EMA), interactive dashboards, and structured AI outputs.
- Deployed production-style architecture via Vite and GitHub Pages.

Remote Factory Digital Twin (Ongoing)

- Developing a smart factory digital twin using large-scale IoT sensor data (1.5M+ records).
- Automated data cleaning and leakage-aware feature engineering (rolling statistics, health indicators).
- Building an initial predictive maintenance pipeline with time-aware RUL modeling and Power BI condition monitoring dashboards.

Music Album Popularity Prediction – ML Research Series

- Evaluated and compared ML models (LR, KNN, SVM, Random Forest, ANN, CatBoost, GNN) with statistical analysis using SPSS.